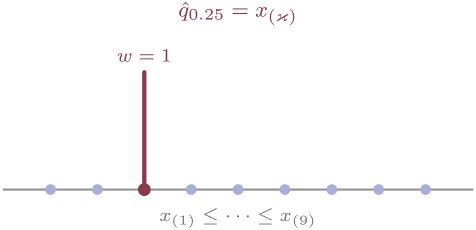
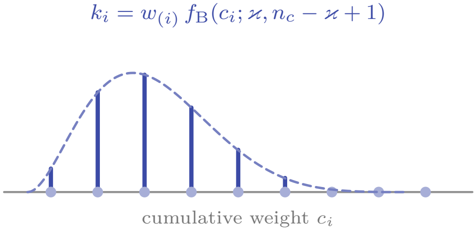


One patient carries the whole gradient. Two patients swap rank and it jumps.



The rank of the $\varkappa$-th of n_c draws is $\mathrm{Beta}(\varkappa, n_c - \varkappa + 1)$ whatever the distribution, so centre $\varkappa/(n_c+1)$ and width $\sqrt{p(1-p)/n_c}$ are set by n_c, not chosen.